Demi Zhang

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EDUCATION

University of Florida, Gainesville, FL Expected May 2028
Ph.D. in Linguistics. Advisor: Edith Kaan.
Ph.D. minor in Computer Science. Advisor: Bonnie J. Dorr.
GPA: 3.96/4.0. First graduate semester at UF: Fall 2023.
Research areas: psycholinguistics, bilingualism, code-switching, EEG/ERP, computational linguistics, multilingual NLP.

Vanderbilt University, Nashville, TN August 2021–May 2023
M.Ed. in English Language Learners, Language Studies and Development track. Advisor: James Booth.
GPA: 3.966/4.0.

Nanjing Agricultural University, Nanjing, China September 2016–June 2020
B.A. in English. GPA: 88.86/100, top 5%.

PUBLICATIONS

- Jian Meng, **Demi Zhang**, Yuling Zhong, Xiaodong Xu, and Edith Kaan. 2026. Comparing Brain Responses to Moral and Semantic Violations. *Brain Sciences*, 16(4), 375. doi:10.3390/brainsci16040375.
- Divyansh Singh, Brodie Mather, **Demi Zhang**, Patrick Lehman, Justin Ho, and Bonnie J. Dorr. 2025. FIDELITY: Fine-grained Interpretable Distillation for Effective Language Insights and Topic Yielding. *Findings of the Association for Computational Linguistics: NAACL 2025*, 2460–2472.
- Michael Bennie, **Demi Zhang**, Bushi Xiao, Jing Cao, Chryseis Xinyi Liu, Jian Meng, and Alayo Tripp. 2025. PANDA: Paired Anti-hate Narratives Dataset from Asia: Using an LLM-as-a-Judge to Create the First Chinese Counterspeech Dataset. *Proceedings of the First Workshop on Multilingual Counterspeech Generation*, 1–12.
- Michael Bennie, Bushi Xiao, Chryseis Xinyi Liu, **Demi Zhang**, Jian Meng, and Alayo Tripp. 2025. CODEOFCONDUCT at Multilingual Counterspeech Generation: A Context-Aware Model for Robust Counterspeech Generation in Low-Resource Languages. *Proceedings of the First Workshop on Multilingual Counterspeech Generation*, 37–46.
- **Demi Zhang**, Bushi Xiao, Chao Gao, Sangpil Youm, and Bonnie J. Dorr. 2024. Modeling Bilingual Sentence Processing: Evaluating RNN and Transformer Architectures for Cross-Language Structural Priming. *Proceedings of the Fourth Workshop on Multilingual Representation Learning*, 127–136.

MANUSCRIPTS AND WORK IN PROGRESS

- **Demi Zhang**, Edith Kaan, and collaborators. Producing code-switches: adaptation of cognitive control in code-switching. Manuscript in preparation.
- Batiah Keissar, Brianna L. Yamasaki, Upasana Nathaniel, **Demi Zhang**, Vedran Dronjic, Bracha Nir, James R. Booth, and Tali Bitan. Neural mechanisms of learning and consolidation of morphologically derived words: the effect of first language similarity. Under review.
- **Demi Zhang**. Predictive processing in naturalistic second-language comprehension through EEG and computational modeling. Dissertation project in development.

RESEARCH EXPERIENCE

Predictive Processing in Naturalistic Language Comprehension

Dissertation-related NSF project 2026–present
University of Florida; collaboration with University of Michigan

- Develop dissertation work connected to the NSF-funded project *Predictive processing in naturalistic language comprehension through EEG and computational modeling*.
- Study bilingual speakers' second-language comprehension using EEG and computational measures of prediction and expectation.
- Focus on speakers of English, Mandarin, and Spanish, and on linking neural responses to model-based estimates of linguistic surprisal and contextual prediction.

Code-switching Production and Cognitive Control

Graduate Researcher July 2024–present
Brain Language and Bilingualism Lab, University of Florida

- Lead a psycholinguistic study examining how code-switching production shapes cognitive control and conflict adaptation.
- Designed the primary E-Prime code-switching production task, bilingual profile questionnaires, verbal fluency task, coding workflow, and statistical analysis framework.
- Created coding schemes for incorrect switching, self-corrections, hesitations, and other production behaviors.
- Manage IRB materials, recruitment documents, protocols, and consent materials.
- Mentor two undergraduate research assistants through weekly meetings, coding training, and project milestones.

Moral Judgment and Semantic Violation Project

Graduate Researcher August 2023–present
Brain Language and Bilingualism Lab, University of Florida

- Designed experimental sentences for a neurolinguistic study of English speakers' brain responses to moral and semantic violations.
- Created recruitment flyers, eligibility questionnaires, and experiment protocols.
- Conduct EEG data collection and model N400 and P600 ERP responses using language-model surprisal measures.
- Co-authored the resulting 2026 *Brain Sciences* article.

Morphological Language Transfer Project

Research Assistant December 2021–May 2023
Brain Development Lab, Vanderbilt University

- Led participant recruitment and data collection for a project on prior knowledge and learning of derivational morphemes in an artificial language.
- Conducted behavioral and fMRI sessions and handled technical setup.
- Analyzed behavioral outcomes including accuracy, reaction time, and individual differences in learning.

Neuro-Cognition of Literacy in Children Who Are Deaf and Hard of Hearing

Research Assistant June 2022–May 2023
Brain Development Lab, Vanderbilt University

- Supported recruitment, data entry, and quality control for a project on neural mechanisms used by higher-skilled deaf and hard-of-hearing readers.

CONFERENCE PRESENTATIONS

- **Demi Zhang**. Predictive processing in naturalistic language comprehension through EEG and computational modeling. MMMM Conference, Nijmegen, Netherlands, June 22–23, 2026.
- **Demi Zhang**. Producing code-switches: Adaptation of cognitive control in code-switching. AMLaP 2025, Prague, Czechia, September 4–6, 2025. Poster.
- Divyansh Singh, Brodie Mather, **Demi Zhang**, Patrick Lehman, Justin Ho, and Bonnie J. Dorr. FIDELITY: Fine-grained Interpretable Distillation for Effective Language Insights and Topic Yielding. NAACL 2025, Albuquerque, NM, April 29–May 4, 2025. Poster.
- Jian Meng, **Demi Zhang**, Yuling Zhong, Xiaodong Xu, and Edith Kaan. Comparing brain responses to moral and semantic violations. Human Sentence Processing 2025, University of Maryland, College Park, March 27–29, 2025. Poster.
- **Demi Zhang**. Producing code-switches: Adaptation of cognitive control in code-switching. Florida Psycholinguistics Meeting, Tallahassee, FL, February 25, 2025. Poster.
- **Demi Zhang**, Bushi Xiao, Chao Gao, Sangpil Youm, and Bonnie J. Dorr. Modeling Bilingual Sentence Processing: Evaluating RNN and Transformer Architectures for Cross-Language Structural Priming. Fourth Workshop on Multilingual Representation Learning, Miami, FL, November 2024.

INVITED TALKS AND GUEST LECTURES

- Bilingualism and second language acquisition. Invited lecture, Anyang Institute of Technology, May 8, 2026.
- Budget traveling. Graduate Linguistics Society hobby talk, University of Florida, March 2, 2026.
- Introduction to Linguistics. Guest lecture, Anyang Normal University, May 23, 2025.

TEACHING EXPERIENCE

University of Florida

ENC 3246: Professional Communication for Engineers Summer A 2026
Instructor of Record

- Taught professional communication for engineering students, emphasizing technical genres, audience analysis, document design, and workplace clarity.

ENC 1102: Argument and Persuasion Summer B 2026
Instructor of Record

- Taught first-year argument and research writing with emphasis on evidence, revision, and public-facing persuasion.

IDS 2935: Language and Computers Spring 2026
Teaching Assistant

- Supported an interdisciplinary course introducing language technologies, computational representations of language, and social questions around language data.

LIN 1140: Language and Emotion Fall 2025
Teaching Assistant

- Supported an undergraduate linguistics course on emotional meaning, expression, and communication.

ENC 2256: Writing in the Disciplines Summer 2024
Instructor of Record

- Taught discipline-based professional writing with attention to genre, audience, and revision.

ENC 3254: Professional Writing in the Discipline Summer 2024
Writing Coach

- Provided individualized feedback, led workshops and office hours, and developed supplementary writing resources.

ENC 1101: Expository and Argumentative Writing August 2023–May 2025
Instructor of Record

- Developed and taught the ENC 1101 curriculum across four semesters, reaching more than 200 undergraduate students.
- Maintained an average instructor evaluation score of 4.72/5.
- Guided students through op-eds, causal analyses, formal proposals, peer review, revision, multimodal communication, and audience adaptation.

New Channel International Education Group, Nanjing, China

ACT Science, AP Calculus, AP Physics, and IELTS Reading June 2020–August 2021
Teacher

- Taught ACT Science at California Crosspoint Academy, Nanjing Branch.
- Taught AP Calculus and AP Physics for students in a Sino-American program at Jinling High School.
- Delivered IELTS reading lectures to undergraduate students at several Nanjing universities.
- Accumulated 1200+ teaching hours and participated in weekly curriculum design meetings.

AWARDS, GRANTS, AND FUNDING

- Grinter Award, University of Florida, $2,500, 2026.
- CLAS travel support for MMMM, University of Florida, $450, 2026.
- Travel support for MMMM, advisor grant, $500, 2026.
- Linguistics Department Travel Grant, University of Florida, $450, 2025.
- CLAS Summer Travel Award, University of Florida, $400, 2025.
- Graduate Student Council International Travel Grant, University of Florida, $500, 2025.

SERVICE, OUTREACH, AND PROFESSIONAL ACTIVITIES

- Brain Awareness Week volunteer teacher: brain puzzle activity, Littlewood Elementary, March 2026.

- Brain Awareness Week volunteer teacher: five senses activity, Westwood Middle School, March 2026.
- Graduate Linguistics Society Seminar Coordinator, University of Florida, Spring 2025; organized seminars and Jolof cook-off.
- Volunteer for linguistics outreach at the Cade Museum, November 15, 2025.
- Brain Awareness Week volunteer teacher: Build a Neuron activity, Glen Springs Elementary, March 25, 2025.
- Volunteer, NAACL 2025, April 2025; supported welcome reception registration and session monitoring.
- Behavioral demonstration for high school students' BLAB tour, February 4, 2025.
- EEG demonstration and training protocol editing for Edith Kaan's course, September 16, 2025.
- Psycholinguistics Reading Group participant; led meetings on July 8, 2025 and October 11, 2025.
- Lunch and campus visit support for faculty candidate Ryan, February 2025.

REVIEWING

- Reviewer, Journal of Neuroscience, 2025.
- Reviewer, ACL Rolling Review, 2025–present.

RELEVANT COURSEWORK

- Linguistics: Phonology 1, Morphology 1, Syntax 1, Introduction to Computational Linguistics, Research Methods in Computational Linguistics, Field Methods, Semantics.
- Computer Science and AI: Large Language Models, Natural Language Processing, Fundamentals of Machine Learning, Research Methods for Human-Centered Computing.
- Statistics and methods: Statistical Inference, Experimental Design, Correlation and Regression.

SKILLS

- Languages: Mandarin Chinese (native), English (proficient), Spanish and French (conversational).
- Programming and tools: Python, R, SQL, E-Prime, HTML, Apache Superset.
- Data analysis: exploratory data analysis, statistical modeling, data visualization, pandas, NumPy, ggplot2.
- NLP and machine learning: tokenization, lemmatization, named entity recognition, sentiment analysis, topic modeling, NLTK, BERT, Hugging Face Transformers, TensorFlow, PyTorch.
- Certificate: NVIDIA Deep Learning Institute Certificate.